UC Berkeley

Department of Electrical Engineering and Computer Sciences

EECS 127 - EECS 227AT. Optimization Models in Engineering

Lecture Note 15

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Theorem 1 (Affine transformation preserves convexity)**.** *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be convex. Consider arbitrary $\mathbf{A} \in \mathbb{R}^{n\times m}$ and $\boldsymbol{b} \in \mathbb{R}^n$. Then $g\colon \mathbb{R}^m \to \mathbb{R}$ defined as $g(\boldsymbol{x}) = f(\mathbf{A}\boldsymbol{x} + \boldsymbol{b})$ is convex on the domain* $\operatorname{dom} g = \{\boldsymbol{x} \in \mathbb{R}^m \mid \mathbf{A}\boldsymbol{x} + \boldsymbol{b} \in \operatorname{dom} f\}$.

Proof. First we prove that $\operatorname{dom} g$ is a convex set. Consider $\boldsymbol{x}, \boldsymbol{y} \in \operatorname{dom} g$, then

$$\mathbf{A}\boldsymbol{x} + \boldsymbol{b} \in \operatorname{dom} f \quad \text{and} \quad \mathbf{A}\boldsymbol{y} + \boldsymbol{b} \in \operatorname{dom} f.$$

For any $0 \le \alpha \le 1$, we have

$$\begin{aligned}
\mathbf{A}\big(\alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{y}\big) + \boldsymbol{b} &= \alpha\mathbf{A}\boldsymbol{x} + (1-\alpha)\mathbf{A}\boldsymbol{y} + \boldsymbol{b} \\
&= \alpha\mathbf{A}\boldsymbol{x} + (1-\alpha)\mathbf{A}\boldsymbol{y} + \alpha\boldsymbol{b} + (1-\alpha)\boldsymbol{b} \\
&= \alpha \underbrace{(\mathbf{A}\boldsymbol{x} + \boldsymbol{b})}_{\in \operatorname{dom} f} + (1-\alpha) \underbrace{(\mathbf{A}\boldsymbol{y} + \boldsymbol{b})}_{\in \operatorname{dom} f} \\
&\in \operatorname{dom} f. \qquad \text{(due to convexity of } f\text{)}
\end{aligned}$$

Therefore, $\alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{y} \in \operatorname{dom} g$, and $\operatorname{dom} g$ is a convex set.
Then, we prove the inequality

$$g\big(\alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{y}\big) \le \alpha g(\boldsymbol{x}) + (1-\alpha)g(\boldsymbol{y})$$

holds for any $\boldsymbol{x}, \boldsymbol{y} \in \operatorname{dom} g$ and any $\alpha \in [0,1]$. In fact, by the convexity of f,

$$\begin{aligned}
g\big(\alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{y}\big) &= f\left(\mathbf{A}\big(\alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{y}\big) + \boldsymbol{b}\right) \\
&= f\big(\alpha(\mathbf{A}\boldsymbol{x} + \boldsymbol{b}) + (1-\alpha)(\mathbf{A}\boldsymbol{y} + \boldsymbol{b})\big) \\
&\le \alpha f(\mathbf{A}\boldsymbol{x} + \boldsymbol{b}) + (1-\alpha) f(\mathbf{A}\boldsymbol{y} + \boldsymbol{b}) \\
&= \alpha g(\boldsymbol{x}) + (1-\alpha) g(\boldsymbol{y}).
\end{aligned}$$

■

Example 1. Let $n = 1$, $m = 3$, $\mathbf{A} = \begin{bmatrix} 5 & -10 & 2 \end{bmatrix}$, $b = -7$. Define $f\colon \mathbb{R} \to \mathbb{R}$ as $f(x) = \mathrm{e}^x$ which is convex. Now, consider $g\colon \mathbb{R}^3 \to \mathbb{R}$ defined as

$$g(\boldsymbol{x}) = f(\mathbf{A}\boldsymbol{x} + b) = f\left(\begin{bmatrix} 5 & -10 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + (-7)\right) = \mathrm{e}^{5x_1 - 10x_2 + 2x_3 - 7}.$$

By Theorem 1, this is also a convex function. ■

Example 2. We prove that $5(x_1 - 10x_2)^4 - 7\log(x_1 - x_2 + 1)$ is convex on the domain $\{\boldsymbol{x} \in \mathbb{R}^2 \mid x_1 - x_2 + 1 > 0\}$:

- Since x^4 is convex, by Theorem 1, $(x_1 - 10x_2)^4$ is also convex.
- Since $-\log x$ is convex, by Theorem 1, $-\log(x_1 - x_2 + 1)$ is also convex.
- Since 5 and 7 are non-negative, the weighted sum of $(x_1 - 10x_2)^4$ and $-\log(x_1 - x_2 + 1)$ with these coefficients preserves convexity (see Theorem 5 in Lecture Note 14).

■

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Definition 1 (Zeroth-order convexity condition)**.** The convexity definition

$$f\big(\alpha \boldsymbol{x} + (1-\alpha)\boldsymbol{y}\big) \le \alpha f(\boldsymbol{x}) + (1-\alpha) f(\boldsymbol{y})$$

is called the *zeroth-order convexity condition* since it doesn't involve any derivatives of f.

Theorem 2 (First-order convexity condition)**.** *Assume* dom f *is a convex and open set (such as* $\mathbb{R}^n$ *or* $(0, +\infty)$*), and* f *is differentiable on the domain. Then* f *is convex if and only if*

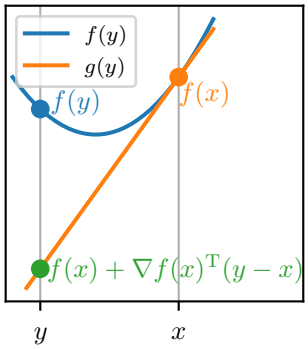
$$f(\boldsymbol{y}) \ge f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^{\mathrm{T}}(\boldsymbol{y} - \boldsymbol{x}), \quad \forall \boldsymbol{x}, \boldsymbol{y} \in \operatorname{dom} f. \tag{1}$$

- **Interpretation 1:** Recall the Taylor expansion:

  $$f(\boldsymbol{y}) = f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^{\mathrm{T}}(\boldsymbol{y} - \boldsymbol{x}) + \text{high-order terms}.$$

  (1) says that the function f is above its first-order Taylor series approximation.

- **Interpretation 2:** $g(\boldsymbol{y}) \coloneqq f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^{\mathrm{T}}(\boldsymbol{y} - \boldsymbol{x})$ is the tangent hyperplane of $f(\cdot)$ at $\boldsymbol{x}$ and is always below the function:



  (Fix $\boldsymbol{x}$, then $f(\boldsymbol{y}) \ge g(\boldsymbol{y})$ for any $\boldsymbol{y}$.)

- We say the tangent hyperplane (e.g., tangential line for $n = 1$) is a *global underestimator* of the function f.

Proof.

- *Necessity proof.* Fix $\boldsymbol{x}, \boldsymbol{y} \in \operatorname{dom} f$. By convexity of f, for any $0 < \alpha < 1$,

$$\begin{aligned}
& f\big(\alpha \boldsymbol{y} + (1-\alpha)\boldsymbol{x}\big) \le \alpha f(\boldsymbol{y}) + (1-\alpha) f(\boldsymbol{x}) \\
\Rightarrow\ & f\big(\boldsymbol{x} + \alpha(\boldsymbol{y} - \boldsymbol{x})\big) \le \alpha f(\boldsymbol{y}) + (1-\alpha) f(\boldsymbol{x}) \\
\Rightarrow\ & f\big(\boldsymbol{x} + \alpha(\boldsymbol{y} - \boldsymbol{x})\big) - f(\boldsymbol{x}) \le \alpha f(\boldsymbol{y}) - \alpha f(\boldsymbol{x}) \\
\Rightarrow\ & \frac{f\big(\boldsymbol{x} + \alpha(\boldsymbol{y} - \boldsymbol{x})\big) - f(\boldsymbol{x})}{\alpha} \le f(\boldsymbol{y}) - f(\boldsymbol{x}).
\end{aligned}$$

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Let $\alpha \to 0$, we get

$$f(\boldsymbol{y}) - f(\boldsymbol{x}) \geq \lim_{\alpha \downarrow 0} \frac{f\big(\boldsymbol{x} + \alpha(\boldsymbol{y} - \boldsymbol{x})\big) - f(\boldsymbol{x})}{\alpha}. \tag{2}$$

Let $g(\alpha) \coloneqq f(\boldsymbol{x} + \alpha(\boldsymbol{y} - \boldsymbol{x}))$. By the chain rule, we have

$$g'(\alpha) = \nabla f\big(\boldsymbol{x} + \alpha(\boldsymbol{y} - \boldsymbol{x})\big)^{\mathrm{T}}(\boldsymbol{y} - \boldsymbol{x}).$$

Substituting this into Equation (2) gives

$$f(\boldsymbol{y}) - f(\boldsymbol{x}) \geq \lim_{\alpha \downarrow 0} \frac{f\big(\boldsymbol{x} + \alpha(\boldsymbol{y} - \boldsymbol{x})\big) - f(\boldsymbol{x})}{\alpha} = g'(0) = \nabla f(\boldsymbol{x})^{\mathrm{T}}(\boldsymbol{y} - \boldsymbol{x}).$$

Therefore, $f(\boldsymbol{y}) \geq f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^{\mathrm{T}}(\boldsymbol{y} - \boldsymbol{x})$.

- *Sufficiency proof.* Fix any $\boldsymbol{x}, \boldsymbol{y} \in \operatorname{dom} f$ and $0 \leq \alpha \leq 1$. Denote $\boldsymbol{z} \coloneqq \alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{y}$. By the first-order convexity condition,

$$f(\boldsymbol{x}) \geq f(\boldsymbol{z}) + \nabla f(\boldsymbol{z})^{\mathrm{T}}(\boldsymbol{x} - \boldsymbol{z}), \tag{3}$$
$$f(\boldsymbol{y}) \geq f(\boldsymbol{z}) + \nabla f(\boldsymbol{z})^{\mathrm{T}}(\boldsymbol{y} - \boldsymbol{z}). \tag{4}$$

Multiply (3) by α, (4) by $1-\alpha$, and add them, we get

$$\begin{aligned}\alpha f(\boldsymbol{x}) + (1-\alpha) f(\boldsymbol{y}) &\geq f(\boldsymbol{z}) + \nabla f(\boldsymbol{z})^{\mathrm{T}}\big(\alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{y} - \boldsymbol{z}\big) \\ &= f(\boldsymbol{z}) = f\big(\alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{y}\big).\end{aligned}$$

Thus f is convex.

■

Theorem 3 (Second-order convexity condition)**.** *Assume* $\operatorname{dom} f$ *is a convex and open set, and* f *is twice differentiable on the domain. Then,* f *is convex if and only if* $\nabla^2 f(\boldsymbol{x}) \succeq \mathbf{0}$, $\forall \boldsymbol{x} \in \operatorname{dom} f$.

Proof. We only give the sufficiency proof. You can refer to https://www.princeton.edu/~aaa/Public/Teaching/ORF523/S16/ORF523_S16_Lec7_gh.pdf for the necessity proof. Consider $\boldsymbol{x}, \boldsymbol{y} \in \operatorname{dom} f$ and assume $\nabla^2 f \succeq \mathbf{0}$ on the domain. By the mid-value theorem (Lemma 1 in Lecture Note 7), there exists a point $\boldsymbol{z}$ on the segment connecting $\boldsymbol{x}$ and $\boldsymbol{y}$ such that

$$f(\boldsymbol{y}) = f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^{\mathrm{T}}(\boldsymbol{y} - \boldsymbol{x}) + \frac{1}{2}(\boldsymbol{y} - \boldsymbol{x})^{\mathrm{T}} \nabla^2 f(\boldsymbol{z})(\boldsymbol{y} - \boldsymbol{x}).$$

Since $\nabla^2 f(\boldsymbol{z})$ is PSD, we have

$$(\boldsymbol{y} - \boldsymbol{x})^{\mathrm{T}} \nabla^2 f(\boldsymbol{z})(\boldsymbol{y} - \boldsymbol{x}) \geq 0.$$

Therefore,

$$f(\boldsymbol{y}) \geq f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^{\mathrm{T}}(\boldsymbol{y} - \boldsymbol{x}),$$

and by the first-order convexity condition, f is convex. ■

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Note: $\nabla^2 f(\boldsymbol{x}) \succ \mathbf{0},\ \forall \boldsymbol{x} \in \operatorname{dom} f$ is a sufficient condition for f to be strictly convex, but it is not a necessary condition.

Example 3. x^4 is strictly convex but $(x^4)''\big|_{x=0} = 0$, so the condition $(x^4)'' > 0,\ \forall x$ doesn't hold. ■

Example 4 (Examples for $n = 1$)**.**

(a) $f(x) = x^2$, then $f''(x) = 2 > 0,\ \forall x$. Thus f is convex.

(b) $f(x) = x^4$, then $f''(x) = 12x^2 \geq 0,\ \forall x$. Thus f is convex.

(c) $f(x) = x^3$, then $f''(x) = 6x \ngeq 0,\ \forall x$. For example, $f''(-1) < 0$. Thus f is non-convex.

(d) $f(x) = \mathrm{e}^{ax}$ where a is arbitrary (positive or negative), then $f''(x) = a^2\mathrm{e}^{ax} \geq 0,\ \forall x$. Thus f is convex.

(e) $f(x) = -\log x$, so $\operatorname{dom} f = (0, +\infty)$ is an open set. We have $f'(x) = -1/x$ and $f''(x) = 1/x^2 > 0,\ \forall x$. Thus, f is convex.

■

Example 5 (Examples for $n = 2$)**.**

(a) $f(\boldsymbol{x}) = x_1^2 + x_2^2 - x_1x_2$, then

$$\nabla^2 f(\boldsymbol{x}) = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \succ \mathbf{0}, \quad \forall \boldsymbol{x}.$$

Thus f is strictly convex.

(b) $f(\boldsymbol{x}) = x_1^2 + x_2^2 - 2x_1x_2$, then

$$\nabla^2 f(\boldsymbol{x}) = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} \succeq \mathbf{0}, \quad \forall \boldsymbol{x}.$$

Thus f is convex.

(c) Let $f(\boldsymbol{x}) = \boldsymbol{x}^{\mathrm{T}}\mathbf{P}\boldsymbol{x} + \boldsymbol{q}^{\mathrm{T}}\boldsymbol{x} + r$ be an arbitrary quadratic function, where $\mathbf{P} \in \mathbb{S}^n$ is a symmetric matrix, $\boldsymbol{q} \in \mathbb{R}^n$, and $r \in \mathbb{R}$. Then $\nabla^2 f(\boldsymbol{x}) = 2\mathbf{P}$ (see Lecture Note 7). Therefore,

(i) f is convex iff $\mathbf{P} \succeq \mathbf{0}$.

(ii) f is strictly convex iff $\mathbf{P} \succ \mathbf{0}$. In fact, unlike the general case where $\nabla^2 f(\boldsymbol{x}) \succ \mathbf{0}$ is only a sufficient (but not necessary) condition for strong convexity, for quadratic functions this condition is also necessary, since the Hessian is constant and fully determines the curvature of f.

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(d) $f(\boldsymbol{x}) = x_1^2 + x_2^2 - 3x_1x_2$, then

$$\nabla^2 f(\boldsymbol{x}) = \begin{bmatrix} 2 & -3 \\ -3 & 2 \end{bmatrix}$$

is sign-indefinite. Thus, f is non-convex.

■

Modern optimization is based on *convex optimization*.

- Consider a general optimization problem with equality and inequality constraints in the following canonical form:

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^n} \quad & f_0(\boldsymbol{x}), \\ \text{s.t.} \quad & f_i(\boldsymbol{x}) \le 0,\ i = 1, \dots, m, \\ & h_j(\boldsymbol{x}) = 0,\ j = 1, \dots, k, \end{aligned}$$

  where:

  (i) The objective function $f_0 \colon \mathbb{R}^n \to \mathbb{R}$ is convex.
  (ii) The functions $f_i \colon \mathbb{R}^n \to \mathbb{R}$ defining inequality constraints are convex.
  (iii) The functions $h_j \colon \mathbb{R}^n \to \mathbb{R}$ defining equality constraints are *affine*.

- The above problem is called a *convex optimization problem*.
- If $m = k = 0$, the problem is just $\min f_0(\boldsymbol{x})$ and is called *unconstrained* convex optimization.
- If $m \neq 0$ or $k \neq 0$, it is called *constrained* convex optimization.

Example 6 (Convex optimization problem)**.** Consider the following optimization problem:

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^3} \quad & \mathrm{e}^{x_1 - x_2} + (2x_1 + x_3)^2, \\ \text{s.t.} \quad & (x_1 + x_2)^6 \le 5x_1 - x_2 + 1, \\ & -1 \le x_3 \le 1, \\ & x_1 + x_2 + x_3 = 5. \end{aligned}$$

Put this into canonical form:

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^3} \quad & \underbrace{\boxed{\mathrm{e}^{x_1 - x_2} + (2x_1 + x_3)^2}}_{f_0(\boldsymbol{x})}, \\ \text{s.t.} \quad & \underbrace{\boxed{(x_1 + x_2)^6 - 5x_1 + x_2 - 1}}_{f_1(\boldsymbol{x})} \le 0, \\ & \underbrace{\boxed{-1 - x_3}}_{f_2(\boldsymbol{x})} \le 0, \\ & \underbrace{\boxed{x_3 - 1}}_{f_3(\boldsymbol{x})} \le 0, \\ & \underbrace{\boxed{x_1 + x_2 + x_3 - 5}}_{h_1(\boldsymbol{x})} = 0. \end{aligned}$$

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Since f_0, f_1, f_2, f_3 are convex functions and h is an affine function, this is a convex optimization problem. ■

- Feasible set:
$$\mathcal{S} = \left\{ \boldsymbol{x} \in \mathbb{R}^n \;\middle|\; \begin{array}{l} f_i(\boldsymbol{x}) \le 0,\ i = 1, \dots, m \\ h_j(\boldsymbol{x}) = 0,\ j = 1, \dots, k \end{array} \right\}.$$
- Why $f_i(\boldsymbol{x})$'s should be convex while $h_j(\boldsymbol{x})$'s should only be affine?

Theorem 4. *The feasible set $\mathcal{S}$ of convex optimization is a convex set.*

Proof. Consider two arbitrary points $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{S}$ and any $\alpha \in [0, 1]$. Let $\boldsymbol{z} = \alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{y}$ be a point on the segment connecting $\boldsymbol{x}$ and $\boldsymbol{y}$. Since f_i is convex,

$$f_i(\boldsymbol{z}) = f_i\big(\alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{y}\big) \le \alpha f_i(\boldsymbol{x}) + (1-\alpha) f_i(\boldsymbol{y}) \le \alpha \cdot 0 + (1-\alpha) \cdot 0 = 0. \tag{5}$$

Since h_j is affine,

$$h_j(\boldsymbol{z}) = h_j\big(\alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{y}\big) = \alpha h_j(\boldsymbol{x}) + (1-\alpha) h_j(\boldsymbol{y}) = \alpha \cdot 0 + (1-\alpha) \cdot 0 = 0. \tag{6}$$

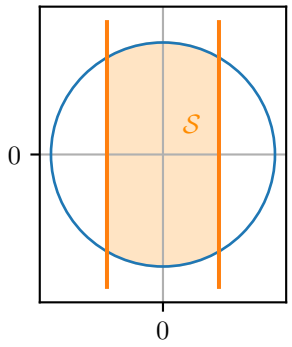
Combining Equation (5) and Equation (6), we have $\boldsymbol{z} \in \mathcal{S}$, thus $\mathcal{S}$ is convex. ■

- Note: Equation (6) requires h_j to be affine and convexity is not enough.
- Note: the difference between linear and affine functions:
  - For a linear function ℓ,
  $$\ell(\alpha\boldsymbol{x} + \beta\boldsymbol{y}) = \alpha\ell(\boldsymbol{x}) + \beta\ell(\boldsymbol{y}), \quad \forall \alpha, \beta.$$
  - For an affine function h, β must be $1-\alpha$; i.e.,
  $$h\big(\alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{y}\big) = \alpha h(\boldsymbol{x}) + (1-\alpha) h(\boldsymbol{y}), \quad \forall \alpha.$$

Example 7 (Feasible set)**.** Consider the following convex optimization:

$$\begin{array}{ll} \min\limits_{\boldsymbol{x} \in \mathbb{R}^2} & \mathrm{e}^{x_1 + x_2} + (x_1 - x_2)^2, \\ \text{s.t.} & x_1^2 + x_2^2 \le 1, \\ & -0.5 \le x_1 \le 0.5. \end{array}$$

The feasible set $\mathcal{S}$ is convex:



■

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